

Reflection of waves and rays

◆ Reflection (waves)

Change of direction that occurs when waves meet a boundary between two media such that the waves return into the medium from which they came.

SYLLABUS CONTENT

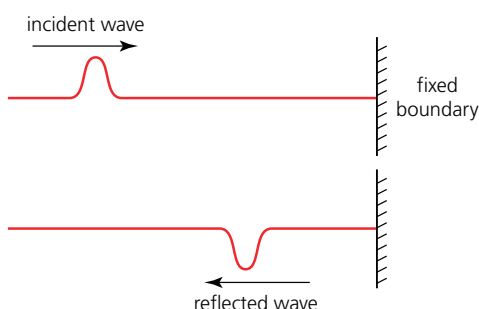
- ▶ Wave behaviour at boundaries in terms of reflection.

When a wave meets a boundary between different media some, or all, of the wave energy will be re-directed back into the first medium. This is called **reflection**.

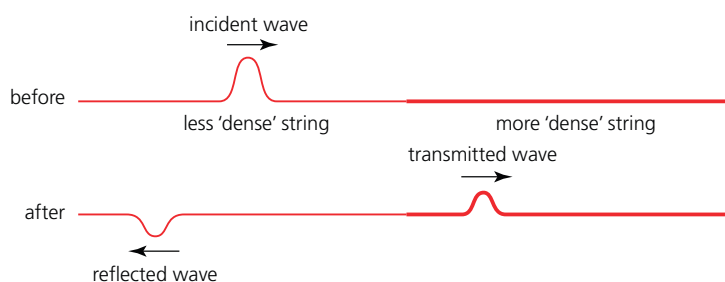
To develop understanding, we will first consider the simplest possible example: a single wave pulse on a rope or string, being reflected at a fixed boundary, as shown in Figure C3.10. A wave travelling towards a boundary is called an *incident wave*. The reflected wave is inverted from this type of boundary: there is a phase change of $\pi/2$.

If the incident waves are continuous, they may combine with the reflected waves to produce a standing wave, as discussed in the next topic, C.4.

Apart from fixed boundaries, where there is no possibility of transmission, waves may also reflect from a boundary where some transmission occurs. Usually, the wave will have different speeds in the two different media.



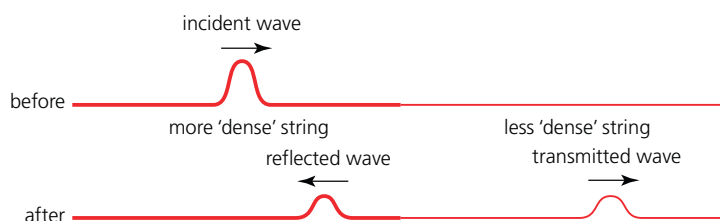
■ **Figure C3.10** Reflection of a pulse off a fixed boundary



■ **Figure C3.11** A pulse travelling into a 'denser' medium

In Figure C3.11 the 'denser' rope has a greater mass per unit length, so that the wave travels more slowly through it. The reflected wave is still inverted. The energy is shared between the reflected wave and the transmitted wave, so that both amplitudes are less than that of the incident wave.

Figure C3.12 shows the situation in which a wave pulse meets a boundary with a medium in which its speed would increase. Note that there is no phase change.



■ **Figure C3.12** Longitudinal waves and pulses behave in a similar way to transverse waves

We will now extend the discussion of reflection to two and three dimensions.



■ **Figure C3.13** Light reflected off and being transmitted by a window

Top tip!

When drawing wavefronts diagrams, make sure that they are *continuous* at the boundaries.

Ray diagrams

Drawings that represent the directions of different waves or particles as they pass through a system.

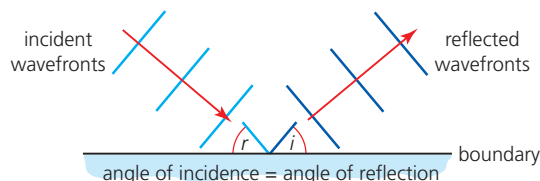
◆ **Incidence, angle** The angle between an incident ray and the normal.

◆ **Angle of reflection (rays)** Angle between a reflected ray and the normal.

◆ **Reflection, law of** Angle of incidence = angle of reflection.

Reflected wavefronts

When *parallel* wavefronts meet a plane (flat) boundary, some or all of them will be reflected so that the angle that the incident wavefront makes with the boundary is equal to the angle that the reflected wavefront makes with the boundary. See Figure C3.14.



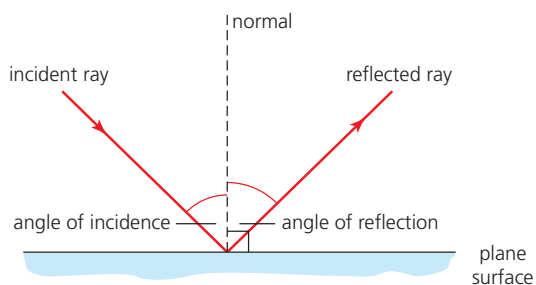
■ **Figure C3.14** Reflection of parallel wavefronts from a straight boundary

When discussing the reflection of light, instead of wavefronts, it is more common to refer to rays, as shown in the **ray diagram** of Figure C3.15. A normal is a line perpendicular to the surface (at the point of incidence). The **angle of incidence** and the **angle of reflection** are measured between the ray and the normal (not the surface).

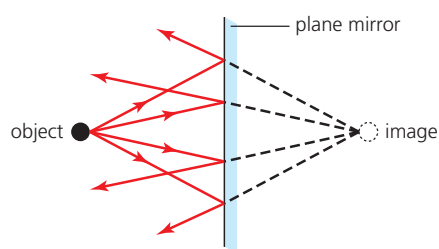
The **law of reflection**:

The angle of incidence equals the angle of reflection.

Figure C3.16 shows radial light rays spreading from a point source (an 'object'). When they strike the plane mirror, the law of reflection can be used to determine the directions of the reflected rays. An eye looking into the mirror will see an image located as far behind the mirror as the object is in front.

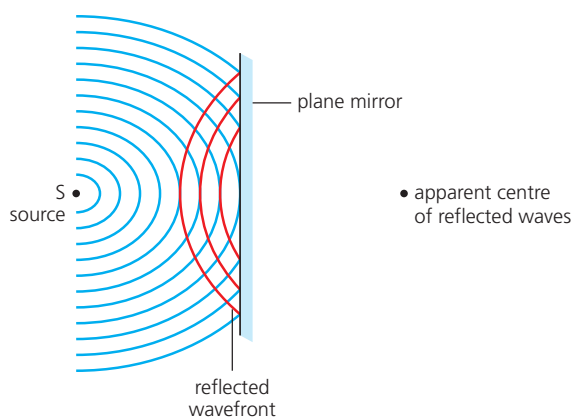


■ **Figure C3.15** Reflection of rays from a plane surface



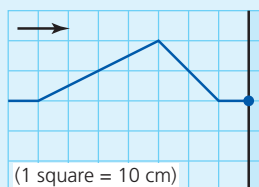
■ **Figure C3.16** Image formed by reflection of rays in a plane mirror

Figure C3.17 represents the same situation using wavefronts instead of rays. The reflected waves appear to come from a point as far behind the reflecting surface as the actual source of the waves (the 'object') is in front.



■ **Figure C3.17** Reflection of circular wavefronts by a plane surface.

- 10 Figure C3.18 shows an idealized pulse on a string approaching a fixed end at a speed of 100 cm s^{-1} . Draw a sketch to show the position of this pulse 0.8 s later.



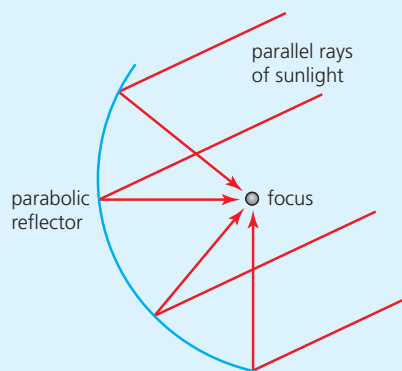
■ **Figure C3.18** An idealized pulse on a string

- 11 Figure C3.19 shows a man looking at a wall. Make a rough copy and add to the wall the smallest mirror that would enable the man to see both the top of his head and his feet. Include light rays that explain your positioning of the mirror.



■ **Figure C3.19** A man looking at a wall

- 12 Predict if there will be a phase change when light waves reflect off a glass surface. Explain your answer.
- 13 Figure C3.20 shows light rays from the Sun being reflected to a focus. Describe the shape of the wavefronts
- arriving from the Sun
 - being reflected to the focus.



■ **Figure C3.20** Light rays from the Sun being reflected to a focus

Inquiry 1: Exploring and designing

Exploring

Sound reflections in large rooms

Sound reflects well off hard surfaces like walls, whereas soft surfaces, such as curtains, carpets, cushions and clothes, tend to absorb sound. A sound that reaches our ears may be quite different from the sound that was emitted from the source because of the many and various reflections it may have undergone. Because of this, singing in the shower will sound very different from singing outdoors or singing in a furnished room. In a large room designed for listening to music (such as an auditorium, Figure C3.21), sounds travel a long way between reflections.



■ **Figure C3.21** A large auditorium designed for effective transfer of sound to the audience

Since it is the reflections that are responsible for most of the absorption of the sound waves, it will take a longer time for a particular sound to fall to a level that we cannot hear. This effect is called *reverberation*. The longer reverberation times of bigger rooms mean that a listener may still be able to hear reverberation from a previous sound at the same time as a new sound is received. That is, there will be some ‘overlapping’ of sounds. Reflections of sounds off the walls, floor and ceiling are also an important factor when music is being produced in a recording studio, although some effects can be added or removed electronically after the original sound has been recorded.

Does your school or college have a performance space, such as a hall or a theatre, where you can make sounds and listen to them carefully when they arrive back at your ears? Or is there a performance space nearby you could visit?

- 1 Research online using search terms such as ‘sound reverberation’ to find out how professional designers adapt and design performance spaces to change reverberation. Then investigate your chosen performance space, inspecting it for installations which affect sound reverberation.
- 2 Discuss and suggest what measurements you could make to test the reverberation in the space.
- 3 Are there any improvements that could be made? State these and explain your reasoning.